Basic Principles of Medicine 2
Module: ER
Lecture No: 21

BPM501 ER 21 Genetic Screening & Counselling

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DLA: Developmental Genetics Objectives

2 Videos are required to be viewed and content included on exam:

1. Developmental Genetics Terminology and Sonic Hedgehog (SHH)
2. Developmental Genetics Hox Genes & Achondroplasia

DLA Developmental Genetics	
SOM.MK.III.BPM2.4.ER.3.GNET.1001	Discuss the vocabulary associated with congenital dysmorphology and give examples.
SOM.MK.III.BPM2.4.ER.3.GNET.1002	Describe the SHH signaling pathway
SOM.MK.III.BPM2.4.ER.3.GNET.1003	Discuss how SHH functions to establish cell fate during development
SOM.MK.III.BPM2.4.ER.3.GNET.1004	Describe disorders associated with defects of SHH signaling
SOM.MK.III.BPM2.4.ER.3.GNET.1005	Describe the HOX gene
SOM.MK.III.BPM2.4.ER.3.GNET.1006	Illustrate the function of the HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1007	Discuss disorders associated with HOX genes
SOM.MK.III.BPM2.4.ER.3.GNET.1008	Describe the general features of achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1009	Discuss the inheritance pattern of the achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1010	Explain the possible outcomes to the offspring of two parents who both have achondroplasia
SOM.MK.III.BPM2.4.ER.3.GNET.1011	Analyze the effect of the FGFR3 G380R mutation and explain why the disorder can be classified as autosomal dominant to gain of function

Lecture Objectives Genetic Screening & Genetic Counseling

SOM.MK.III.BPM2.4.ER.3.GNET.1012	Differentiate between population vs. targeted screening
SOM.MK.III.BPM2.4.ER.3.GNET.1013	Discuss the difference between predictive vs. diagnostic tests
SOM.MK.III.BPM2.4.ER.3.GNET.1014	Discuss the significance of timing when utilizing markers for predictive testing
SOM.MK.III.BPM2.4.ER.3.GNET.1015	Evaluate some existing targeted screening programs for groups of people at high risk for certain genetic diseases: PKU; Thalassemia; Sickle cell disease; Cystic fibrosis
SOM.MK.III.BPM2.4.ER.3.GNET.1016	Summarize the requirements of a genetic screening program
SOM.MK.III.BPM2.4.ER.3.GNET.1017	Define “genetic counselling” and identify the founding principles
SOM.MK.III.BPM2.4.ER.3.GNET.1018	Define the role of a genetic counselor in the face of a positive diagnosis: accuracy of the tests, calculation of reoccurrence risk
SOM.MK.III.BPM2.4.ER.3.GNET.1019	Appreciate the need to provide appropriate assistance in the decision-making process of patients when faced with a positive result and further testing

Recommended Reading

- **Reading suggestion**: Korf & Iron 4th edition: Case studies that are found in chapters 11, 13 & 15
- **Required reading**: Korf & Irons: Hot Topic 7.1 – Carrier Testing – p135
- **Required reading:** See figure 6.19, and accompanying text – p113

Before we get started….

- We will be speaking about genetic testing, presenting results to families, and the choices the families may make
- During this discussion, we will touch on issues that may be sensitive to many of us
- We will endeavor to respect all opinions and beliefs, discussing clinically relevant choices that families may consider when faced with the potential of having a child with a genetic difficulty

Clicker Question Next!

Genetic screening

Targeted Screening

- Screening of populations known to be at risk:
  - Affected individual in the family
    - Reduced penetrance or late onset
  - Screening of high-risk ethnic groups
    - Persons at risk of having children with a genetic disease **(carriers)**

Population Screening

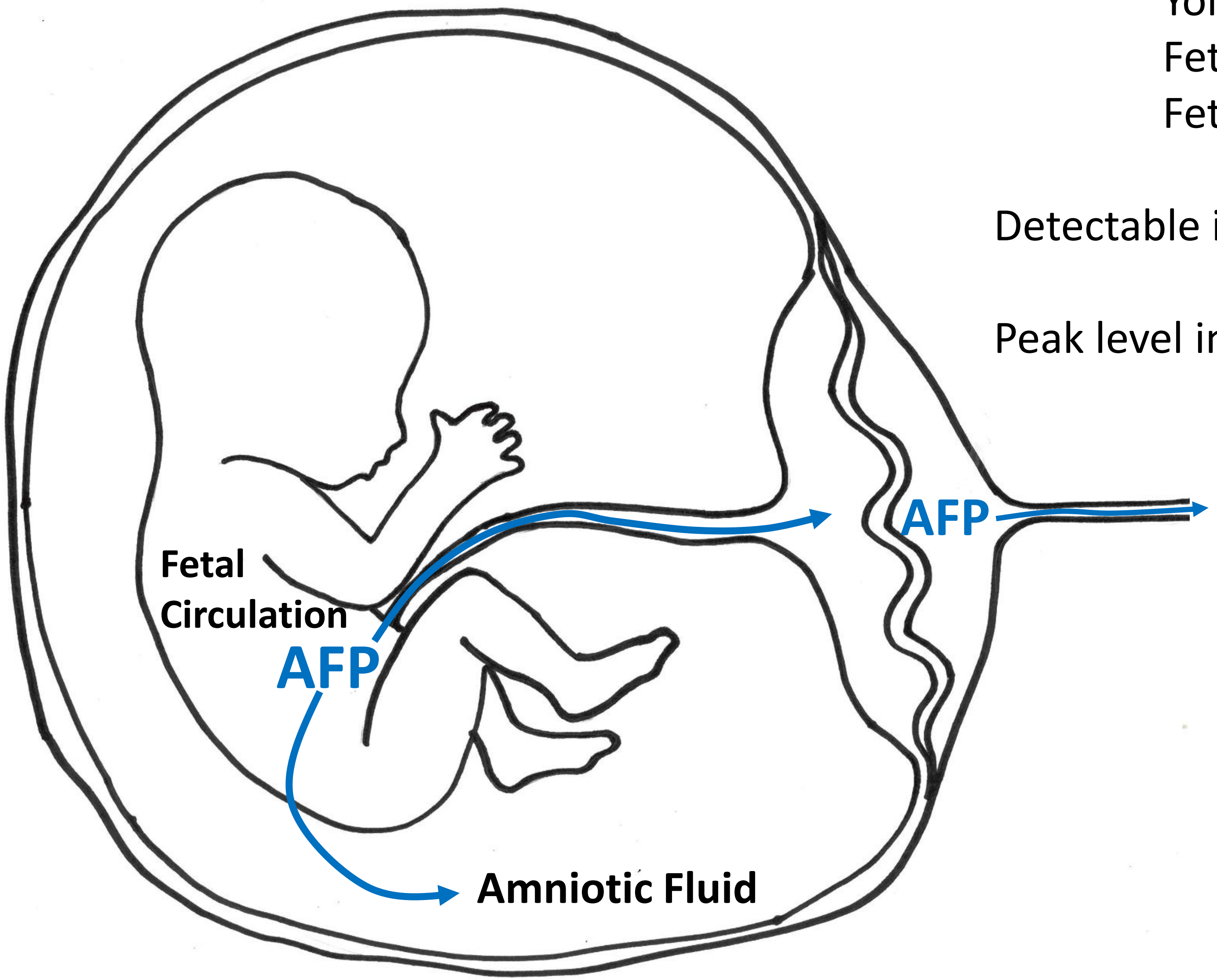
- Screening all members of a designated population regardless of family history
  - PAP, breast, prostate
  - Prenatal screening and Newborn screening

Purpose of Prenatal Screening

Typically used to detect 3 abnormalities

- **Trisomy 21** (Down Syndrome) ~1/830 live births
- **Trisomy 18** ~1/7,500 live births
  - *Trisomy 13 has similar results to 18 but is ~1/22,700 live births*
- **Neural tube defects** ~1/2,000 live births

AFP from Fetal to Maternal Serum



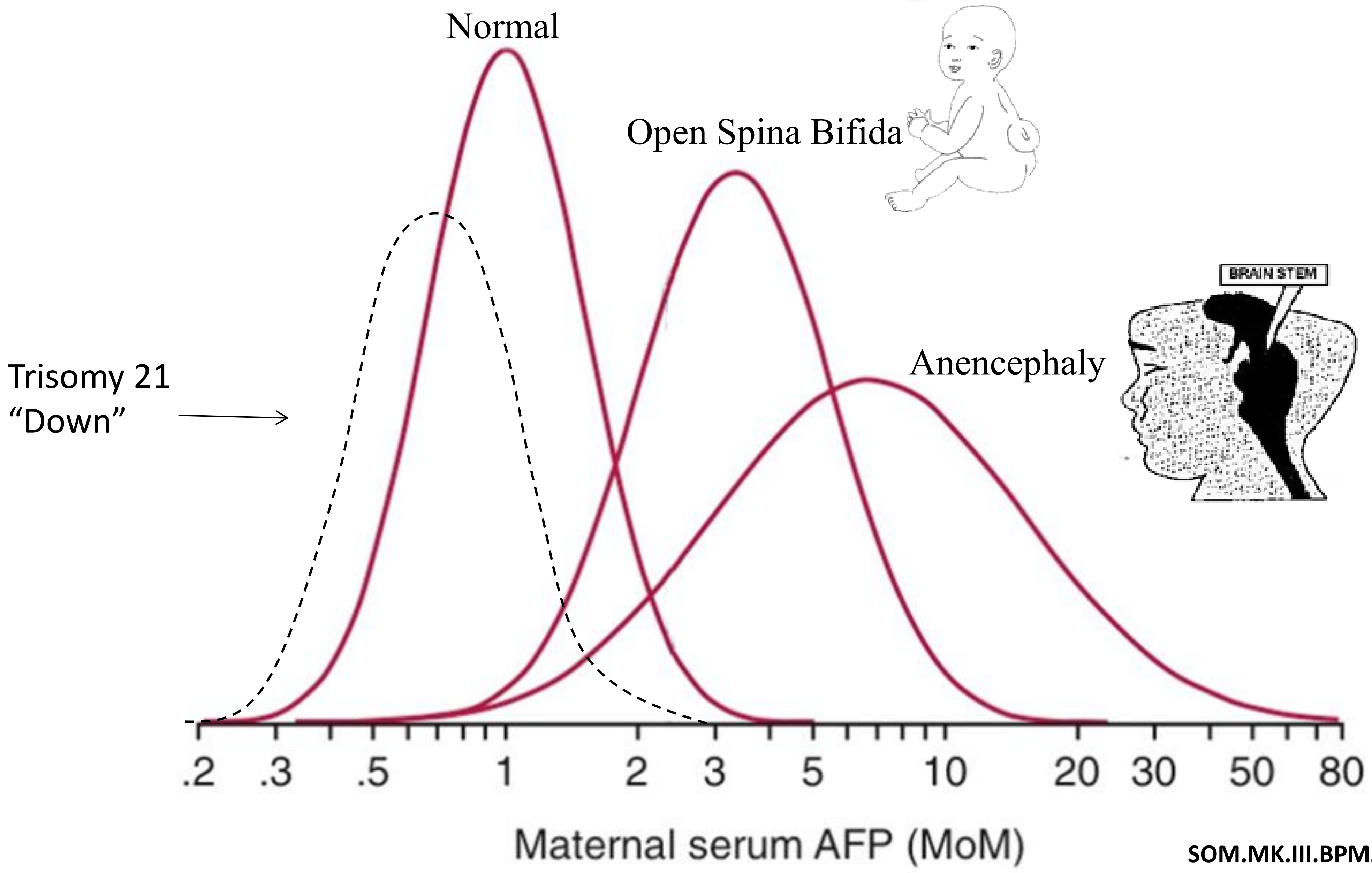
Alpha Fetoprotein (AFP) is synthesized in:

- Yolk sac
- Fetal GI tract
- Fetal liver

Detectable in fetal serum at 6 weeks

Peak level in 12-14 weeks

AFP Serum Screening as a Marker



Prenatal Screening

Non-invasive	Optimal time	
Maternal Serum Screening	16 weeks	Predictive
First trimester screen	11-14 weeks + 1 day	
Second trimester screen	14 weeks + 2 days-20 weeks	
Ultrasound (fetal anomaly scan)	18 weeks	
Nuchal translucency	11-14 weeks	
Invasive		
Chorionic villus sampling	10-14 weeks	Diagnostic
Amniocentesis	15-18 weeks	

Maternal Serum Screening Markers

First Trimester Tests: 11-14 weeks + 1 day

Pregnancy associated plasma protein-A (PAPP-A)
Human chorionic gonadotropin (β-hCG)

Second trimester tests: 14 weeks + 2 days to 20 weeks

Triple Test: looks for 3 markers

AFP (alpha fetoprotein)
Estriol (unconjugated estriol, or uE_3)
Human chorionic gonadotropin (β-hCG)

Quad test: Includes another marker

Inhibin A

Maternal Serum Screening Markers

	First Trimester Screen			Second Trimester Screen			
	PAPP-A	Free β-hCG	Nuchal Translucency	uE3	AFP	Free β-hCG	Inhibin A
Trisomy 21	↓	↑	↑	↓	↓	↑	↑
Trisomy 18	↓	↓	↑	↓	↓	↓	N-↓
Trisomy 13	↓	↓	↑	N-↓	N-↓	N-↓	N
Neural Tube Defect					↑		

AFP, alpha-fetoprotein;
β-hCG, human chorionic gonadotropin β subunit;
PAPP-A, pregnancy associated plasma protein A
uE_3, unconjugated estriol
N, Normal levels

Maternal Serum Screening Patterns

- Low serum AFP, estriol and PAPP-A with high β-hCG and Inhibin A indicate a risk for Trisomy 21 (Down Syndrome)
- Low serum AFP, estriol, PAPP-A and β-hCG and indicate a risk for trisomy 18 (Edward syndrome)
- Trisomy 13 (Patau syndrome, most rare) first trimester test often shows reduced PAPP-A and β-hCG, however these reductions tend towards normal by the second trimester, thus second trimester test is uninformative for trisomy 13

Non-invasive Prenatal Screening (NIPS or NIPT)

- **Detects Cell Free DNA in Maternal Serum, 10 weeks till delivery**
  - cell-free DNA (cfDNA) from the fetus in the maternal circulation
    - Fetal fraction (amount of fetal DNA in maternal circulation)
    - Small segments of DNA 25-30 bp
  - Used primarily for detection of aneuploidy
    - Trisomy 13, 18, 21 or extra or missing X and Y
  - May be used for detecting mutations in specific genes

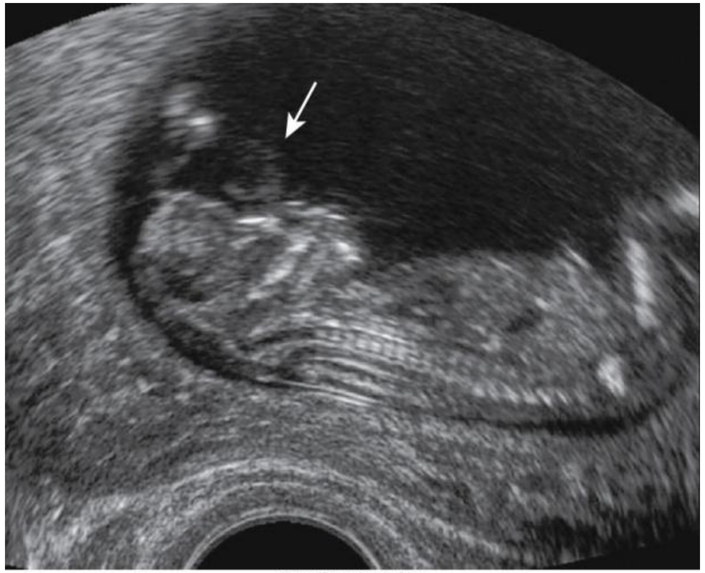
- **This is a screening test that predicts risk for an affected fetus**
  - Analysis of fetal DNA in maternal blood (non-invasive) has the advantages of earlier detection of trisomy and eliminates the risk of spontaneous abortion by amniocentesis or CV sampling
  - Requires follow-up prenatal testing to confirm any findings

- **Challenges/complications in testing**
  - Young maternal age
  - Twin or other multiples
  - Previous pregnancies

Ultrasonography

At 18 weeks can detect many different structural abnormalities

Neural tube defects are best detected by ↑maternal AFP combined with **ultrasound**

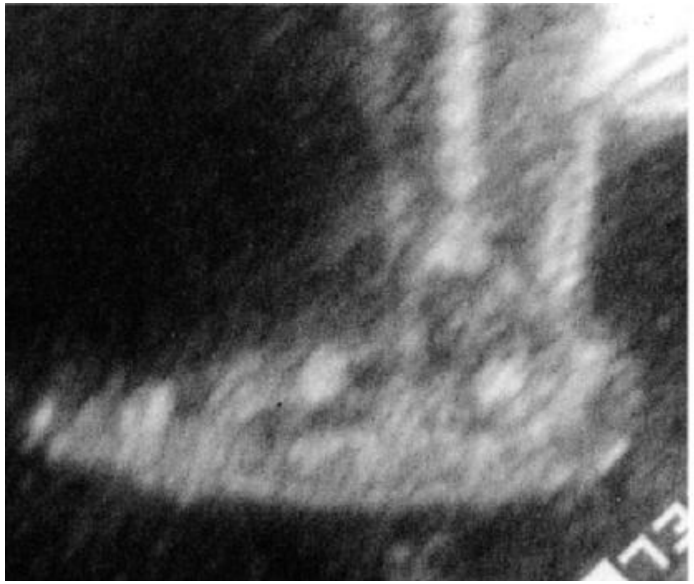


(Courtesy of Dr Helen Liversedge, Exeter.)

Anencephaly

(Neural tube defect)

Emery's fig 21.7



Rocker-bottom foot

Edwards Syndrome

(trisomy 18)

Structural defect

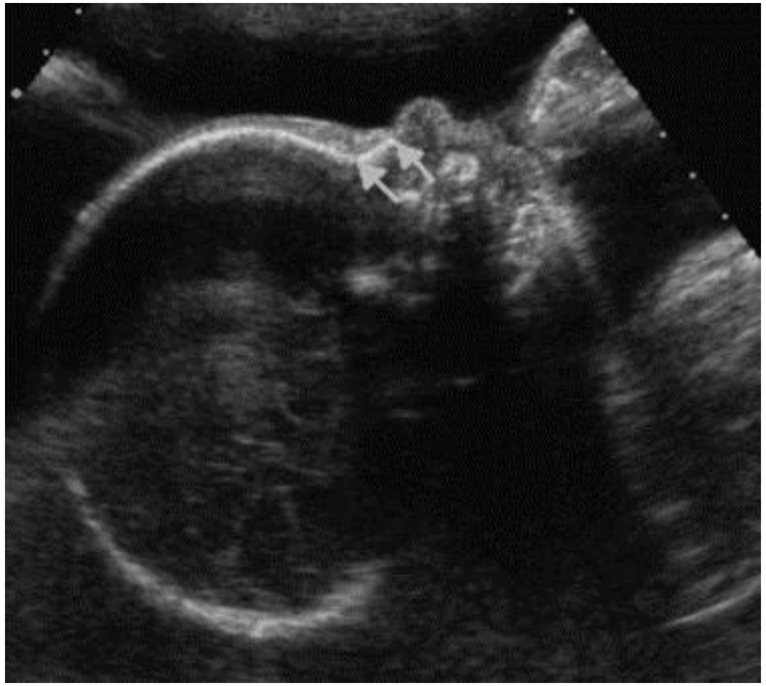
micrognathia

(Cri-du-chat)

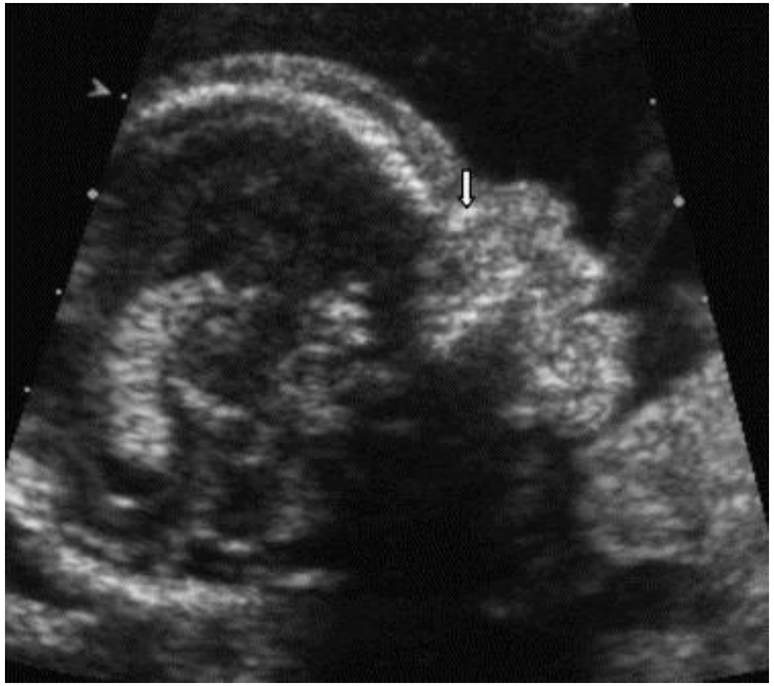
Emery's fig 21.4

Ultrasonography

Nasal Bone, hypoplastic or absent may suggest Trisomy 21



Second trimester, normal

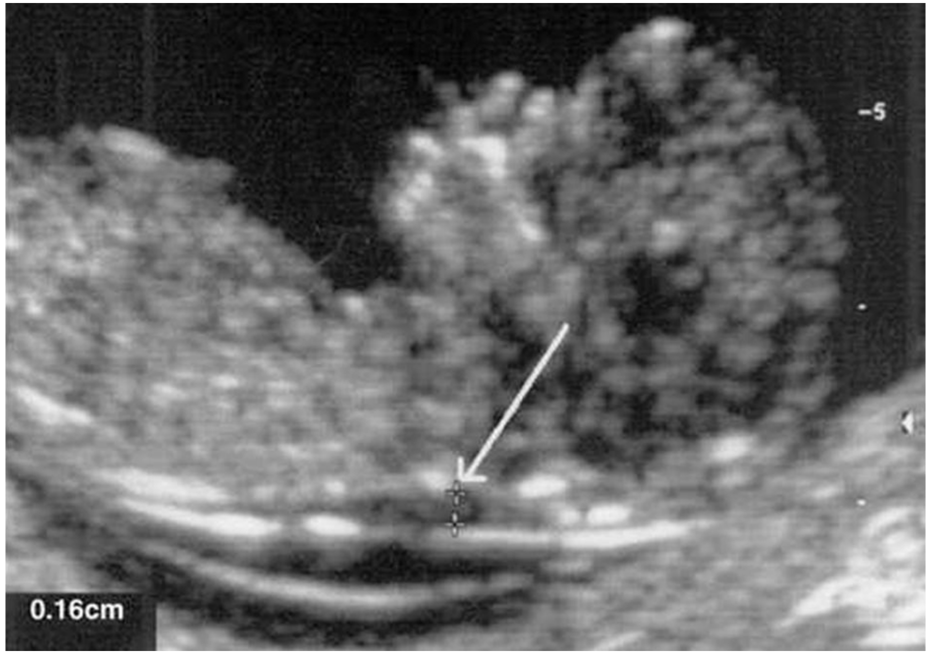


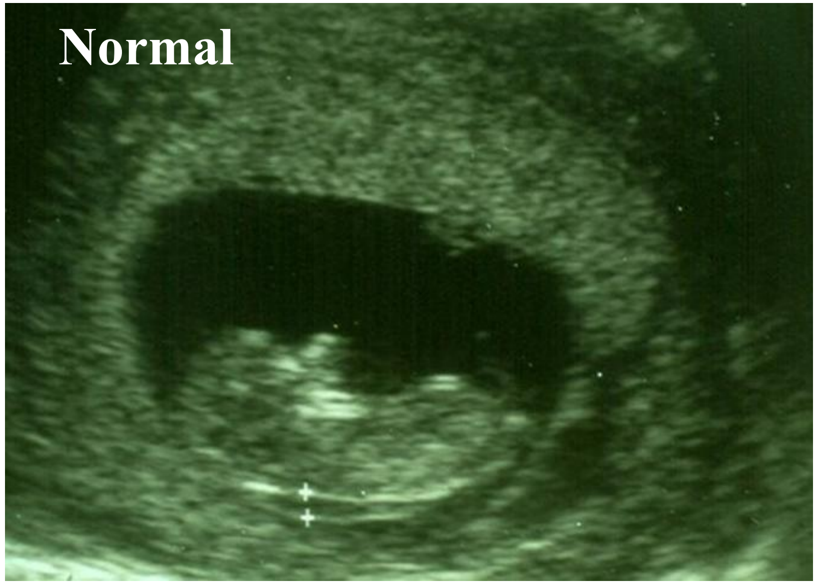
Second trimester, hypoplastic, T21

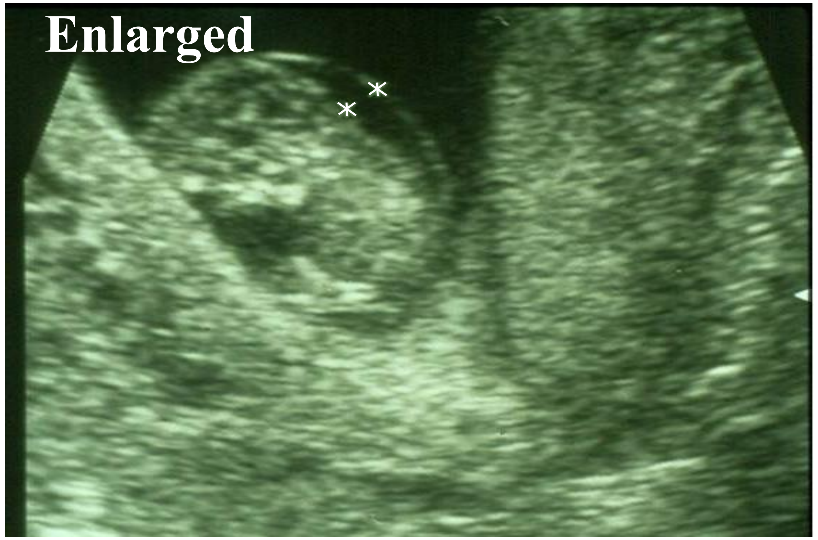
Second trimester, absence, T21

Bouley R, Sonek J. Fetal nasal bone: the technique. Down's Screening News 2003;10:33-4.

Enlarged Nuchal Translucency (NT) 11-14 weeks







Increased NT thickness is <u>associated</u> with chromosome aneuploidy

- Trisomies 21, 18, 13,
- Triploidy
- Turner syndrome (45,X)

If screening shows an increased probability of an affected pregnancy, the family may choose to undergo an invasive diagnostic test

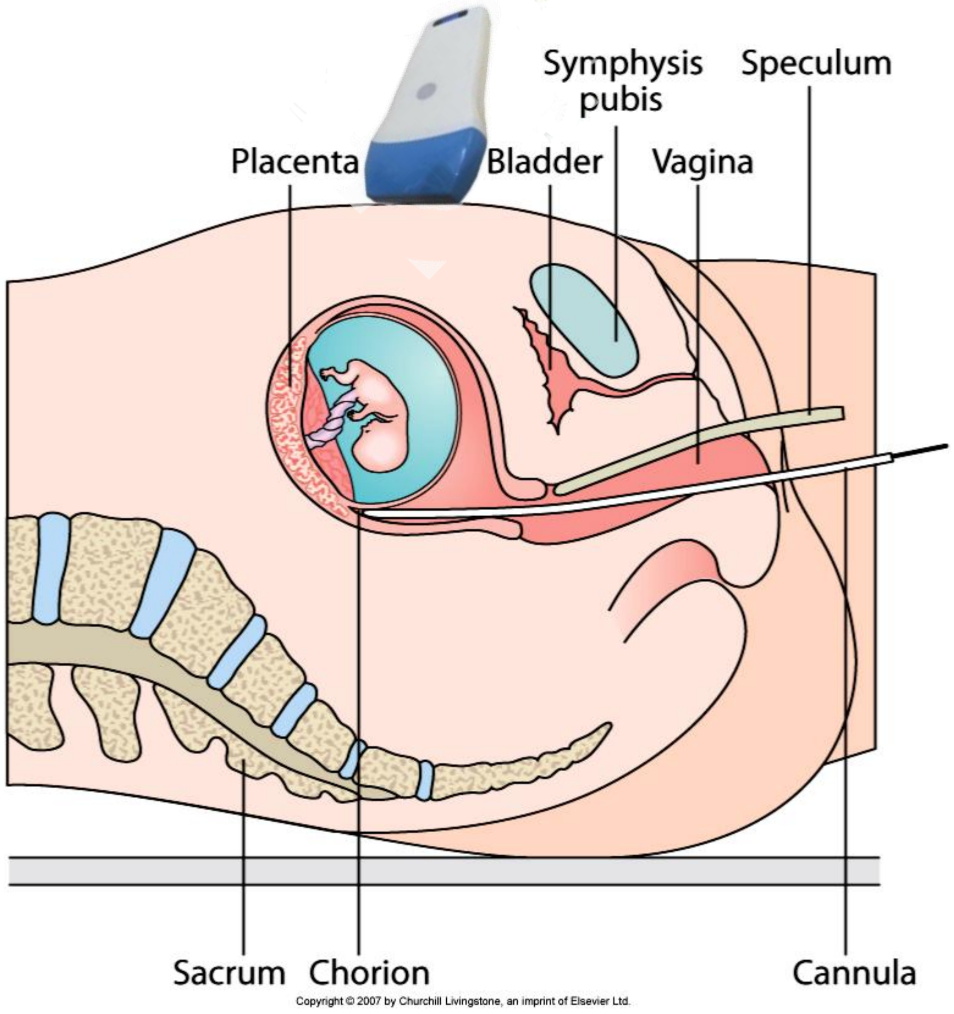
Risk of inducing a miscarriage

Maternal Serum Screening Prenatal Genetic Counseling

- Prior to screening, a Genetic Counselor will inform the family that this is a SCREENING and not a diagnostic test
- Informed consent will force couples to consider their fetus may have a birth defect
- A positive screening result may require invasive testing
- Information comes either just before or in the second trimester when family bonding has begun
- Inherent risk the invasive procedure may harm the pregnancy
- Issue of considering pregnancy termination if an abnormality is detected
- Conflict over additional bonding with fetus if an abnormality is detected

Chorionic Villus Sampling

- 10-14 weeks
- Involves removal of fetal cells by aspiration from the inner surface of placenta
- Direct chromosomal analysis by FISH allows rapid results



Emery's fig. 21.2

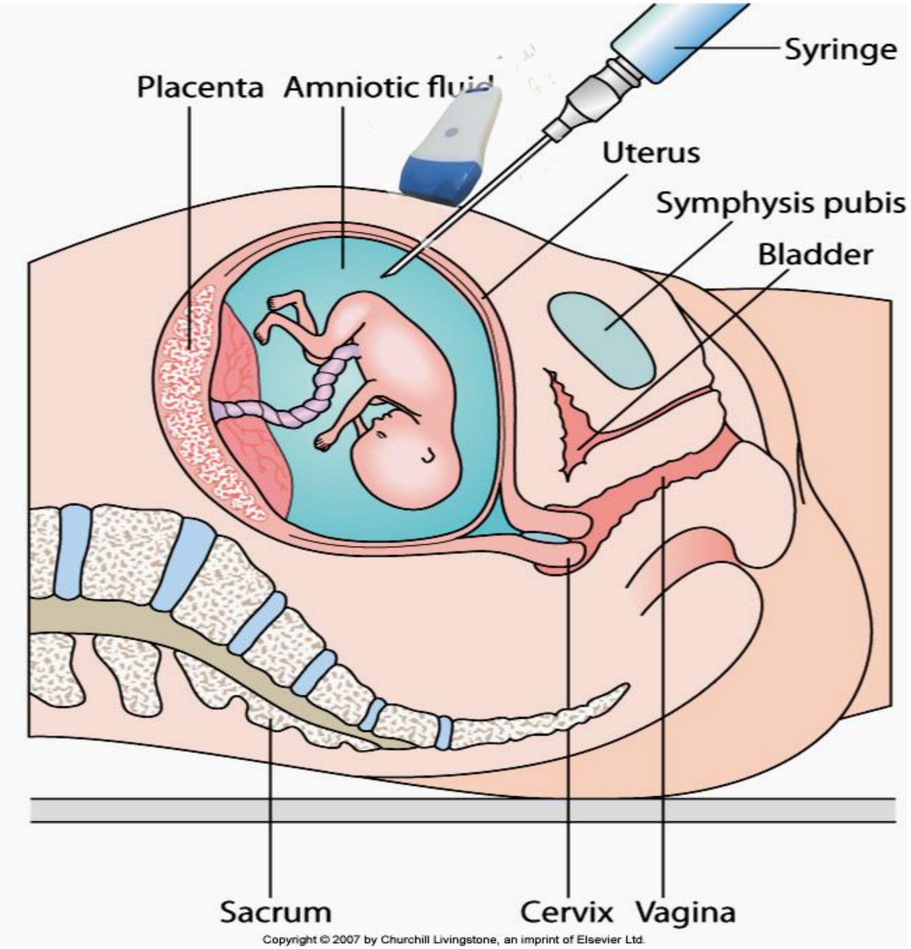
Amniocentesis

15-18 weeks

10-20 ml of amniotic fluid is aspirated trans-abdominally guided by ultrasound

- Fetal cells are pelleted by centrifugation and used for chromosome analysis

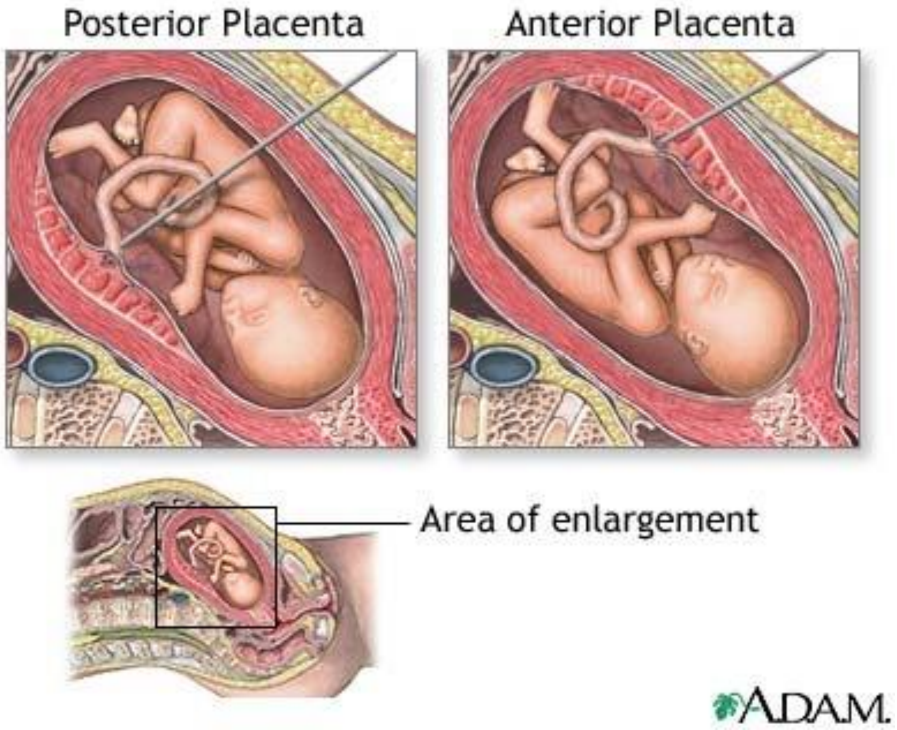
-supernatant can be used for AFP assay

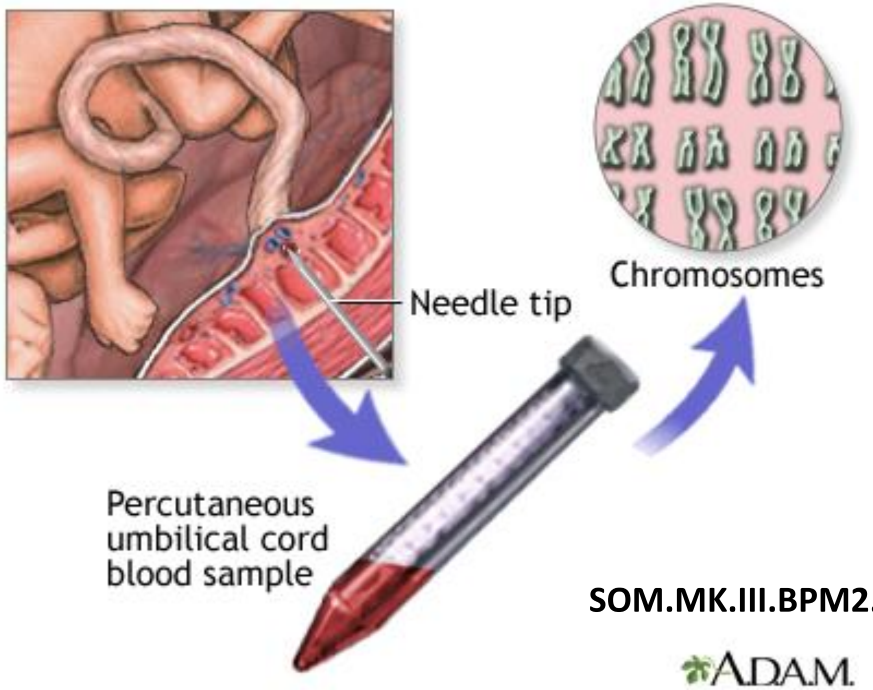


Emery's fig. 21.1

Percutaneous umbilical blood sampling PUBS

- Usually done after 18 weeks of gestation
- Performed when there is a delayed suspicion of a chromosomal abnormality usually detected by ultrasound in 2nd trimester





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Family with History of Genetic Disease or Abnormal Screening Results

Prenatal Genetic Counseling for Maternal Screening

Prenatal genetic counselor involvement during pregnancy

- Abnormal serum screening (neural tube defects or chromosome abnormalities)
- Abnormal ultrasound finding
- Positive invasive diagnostic testing

Prenatal genetic counselor involvement prior to conception

- History of infertility, miscarriages or stillbirths
- Couples older than 35 years
- Previous child with chromosome abnormality or other genetic disorder
- Previous child with birth defect (neural tube defect) or multiple congenital anomalies
- Specific ethnicity that may have a higher incidence of certain disorders

What is available for a family with a history of Genetic Disorder

Preimplantation Genetics

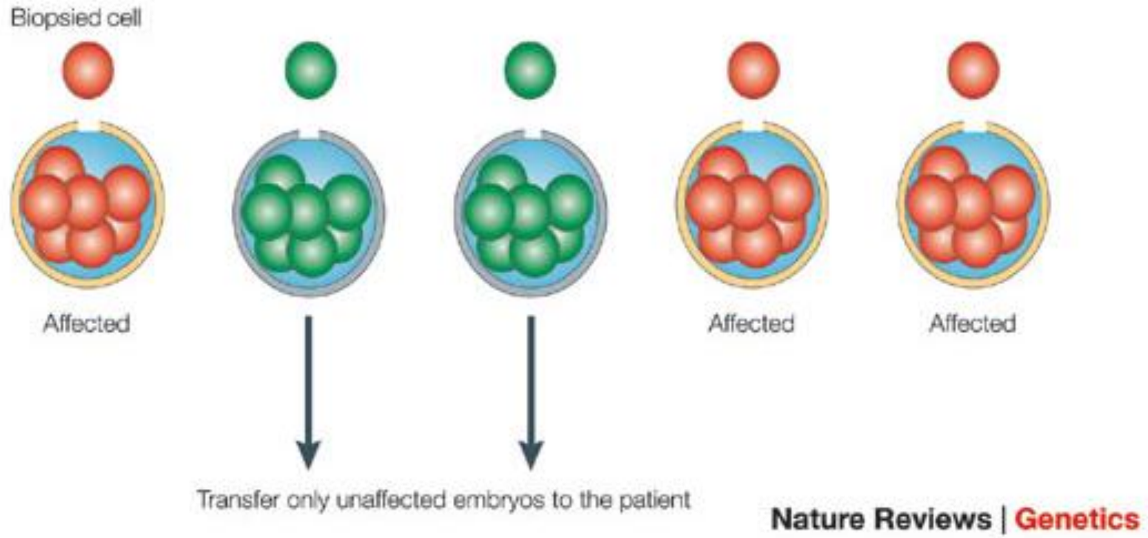
- Preimplantation genetics involves using **artificial reproductive technology (ART)**.
- This involves fertilizing the egg with sperm by in vitro fertilization and testing the embryo for a specific condition **before** it is implanted into the mother

polar body after fertilization

blastocyst; 8-16 cell stage (Day 3)

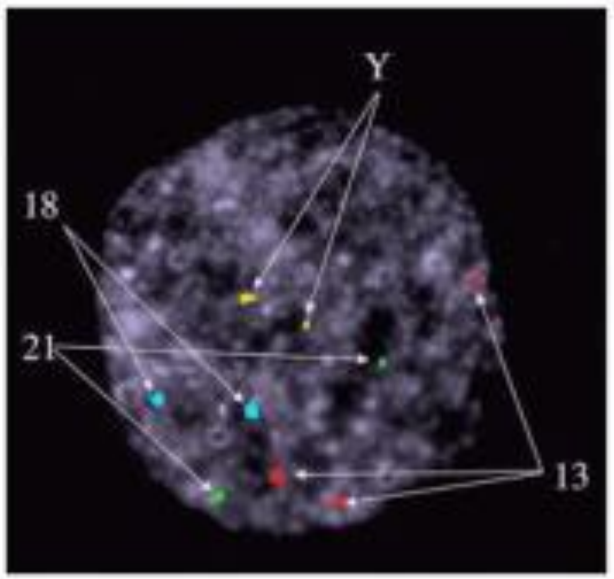
Nature Reviews | Genetics

Preimplantation Genetics



By Day 5

Testing
PCR
FISH
NGS



Multiple FISH demonstrating chromosomes 13 (triploid), 18, 21, and XX

What is genetic counseling???

- **Non-directive Counseling**
- Genetic counseling is the process of helping people **understand** and **adapt** to the medical, psychological and familial implications of genetic contributions to disease.
- **This process integrates the following**:
  - Discuss the cause of the family's disease
  - Interpretation of family and medical histories to assess the chance of **disease occurrence or recurrence** (in offspring)
  - Education about **inheritance, testing, management**, prevention, resources and research.
  - Counseling to promote **informed choices** and adaptations to the risk or condition
  - Support resources
  - Help treat psychosocial issues:
    - Guilt, fear, shame, grief, search for meaning

Newborn screening

Criteria for Undertaking Neonatal Screening

- The disorder produces irreversible damage if untreated early in life
- there is a treatment available for the disorder
- early intervention is effective
- a reliable lab test for detection
  (A *positive result should be confirmed immediately by retesting*)

A BILL

To amend the Public Health Service Act to establish grant programs to provide for education and outreach on newborn screening and coordinated followup care once newborn screening has been conducted, to reauthorize programs under part A of title XI of such Act, and for other purposes.

Be it enacted by the Senate and House of Representatives of the United States of America in Congress assembled,

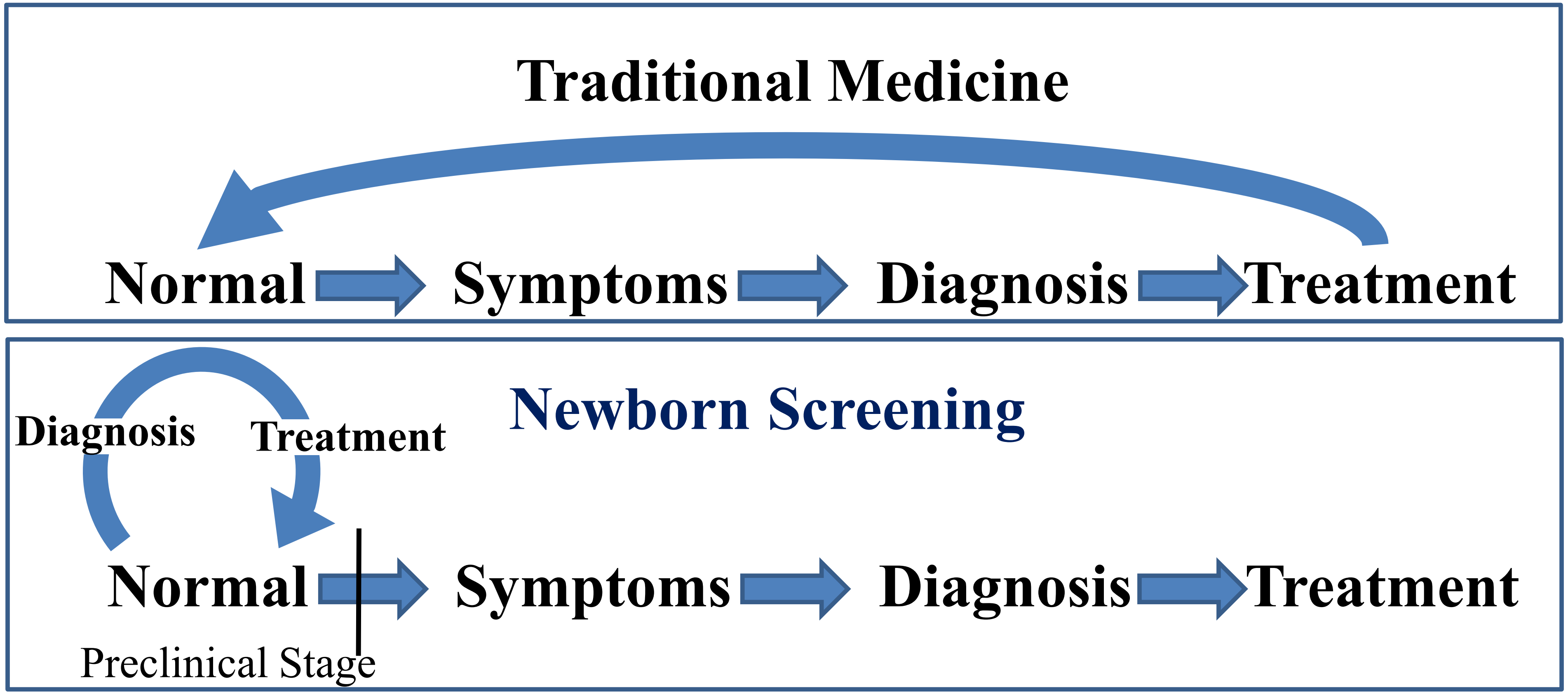
SECTION 1. SHORT TITLE.

This Act may be cited as the "Newborn Screening Saves Lives Act of 2008".

SEC. 2. IMPROVED NEWBORN AND CHILD SCREENING FOR HERITABLE DISORDER.

Newborn screening

- Traditional Testing: based on signs, symptoms, or family history
- NBS: done on all to identify those children to treat
- Preclinical: before symptoms, disease or death



Phenylketonuria - First to be Screened

- Newborn screening originally by a bacterial inhibition test that detects elevated levels of phenylalanine in circulation
- Today, positive newborn screening test has to be immediately confirmed by elevated Phe levels in circulation by a more specific assay (Quantitative mass spectrometry)
- Management involves the lifelong restriction of dietary Phe, that results in a marked improvement and normal mental development in the child

Not treated

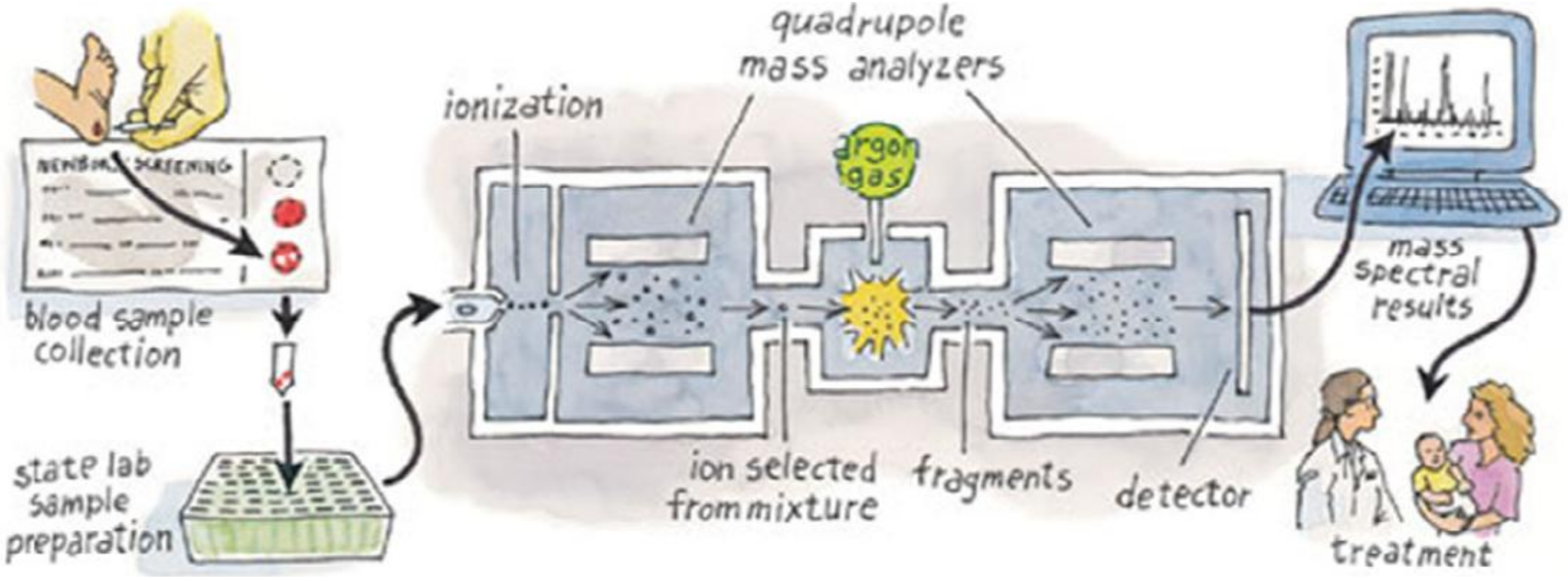
Treated from 2 y, MR, can not talk

Treated from birth, normal development

Newborn screening – spots of blood

2 MS/MS systems
~120,000 samples/yr at the Mayo Clinic
Rochester, Minnesota

PKU Newborn Screening by Tandem MS (MS/MS)



- Molecules (amino acids) are ionized & injected into mass analyzer of MS/MS
- Selected ions are fragmented
- Fragments separated by second mass analyzer & amino acids are detected based on their signature fragment pattern

Disorder
Amino acid disorders
Phenylketonuria (includes clinically significant hyperphenylalaninemia variants)
Maple syrup urine disease
Homocystinuria
Citrullinemia I
Argininosuccinic acidemia
Organic acid metabolism disorders
Isovaleric acidemia
Glutaric acidemia type I
Hydroxymethylglutaric aciduria
Multiple carboxylase deficiency
Methylmalonic acidemia (mutase deficiency)
Methylmalonic acidemia CblA,B
3-Methylcrotonyl-CoA carboxylase deficiency
Propionic acidemia
Beta-ketothiolase deficiency
Fatty acid oxidation disorders
Medium-chain acyl-CoA dehydrogenase deficiency
Very long-chain acyl-CoA dehydrogenase deficiency
Long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency
Trifunctional protein deficiency
Carnitine uptake defect
Hemoglobinopathies**
Hb SS
Hb SC
Hb S/β thalassemia
Other disorders
Primary congenital hypothyroidism (excluding secondary, transient, or other)
Biotinidase deficiency (including partial)
Congenital adrenal hyperplasia (excluding non 21-hydroxylase deficiency)
Classical galactosemia+variant (excluding GALK and GALE)
Cystic fibrosis (including nonclassical)

The first test, Guthrie bacterial inhibition assay, now:

tandem mass spectroscopy

In the DM module you will be learning about protein, lipid and carbohydrate metabolism. Deficient enzymes in these pathways lead to most of the disorders presented in this lecture

electrophoresis

Immunoreactive trypsin

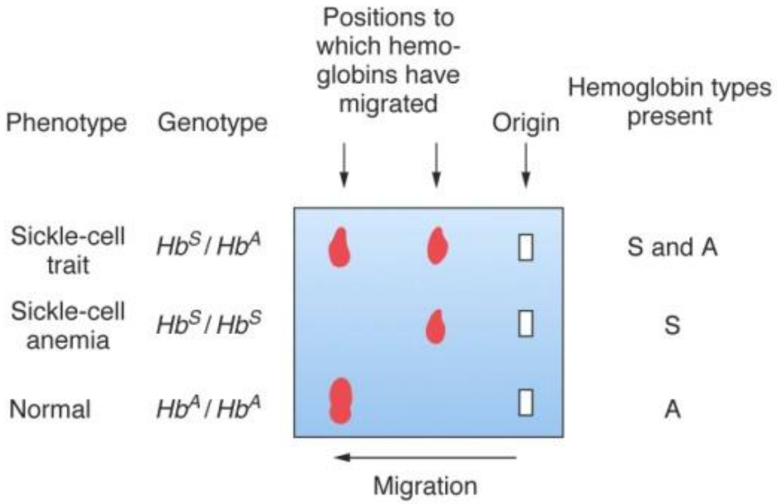
Sickle Cell Disease

Point mutation in β-globin resulting in anemia

Detection

- Hemoglobin electrophoresis
- DNA test:

  -Southern blot, PCR-RFLP, ASO test



GAG to GTG
Glu to Val

Treatment

- Prevention of sickling crisis, blood transfusions, use of the drug hydroxyurea

(reactivating fetal hemoglobin)

Thalassemia (α and β)

- Both show autosomal recessive inheritance
- Both can be detected by mean corpuscular hemoglobin and hemoglobin electrophoresis
- Screening of young adults for the β-thalassemia carrier status has markedly reduced the birth incidence of β-thalassemia in Cyprus, Greece

Cystic Fibrosis

Mutation in the CF transmembrane conductance regulator (CFTR)

FYI- Incidence: 1 in 3,000 Caucasians or Ashkenazi Jewish, 1 in 8,000 Hispanics, 1 in 15,000 African Americans, 1 in 32,000 Asians.

Detection

- **Immuno-reactive trypsinogen** (IRT) in blood followed by a DNA test
- **DNA** looking for ~32 of the most common mutations
- **Sweat chloride test** (confirmatory test)

Management

- Antibiotics to combat respiratory infections, gene therapy, management of associated malabsorption

SCID- Severe Combined Immunodeficiency

SCID is the result of genetic defects which impair the normal development of T-cells

(x-linked recessive and autosomal recessive)

SCID infants are phenotypically normal but acquire life-threatening infections within a few months of life

Detection

- Healthy T lymphocyte development will show presence of the by-product T cell receptor excision circles (TRECs)
- TRECs are normally made in large numbers, while barely detectable in infants with SCID
- TRECs are quantitated by real-time quantitative polymerase chain reaction (RT-qPCR) on DNA extracted from routine NBS cards

Management

- bone marrow transplant

SOM.MK.III.BPM2.4.ER.3.GNET.1014&15

NBS and the Genetic counselor

- “My first patient is a 4-week-old girl coming in due to a **positive newborn screen** for cystic fibrosis (CF). Before coming to see me this afternoon, she will have a **sweat test.”**
- She has no medical history for **respiratory or gastrointestinal** problems. Upon her arrival I learn that she has had a **negative sweat test** and may be a **carrier of cystic fibrosis.**
- This is the first thing I explain to the parents, adding that being a carrier does not have any medical implications for their daughter. They look very relieved.
- During the remainder of the session, we talk about how CF is inherited in an **autosomal recessive** fashion, and most likely that one of the parents is a CF **carrier.**
- In addition, when their **daughter is planning her family**, she may have a similar consultation with genetics.
- Finally, as it is possible that both parents could be carriers but only one passed on the mutant CF gene, **I suggest that both parents be tested for carrier status**. The parents opt to come back to have their carrier status tested and an appointment is made for next week.

Carrier screening for Parents with Affected Child

- Most couples learn that they are carriers only after the birth of an affected child
- Carrier screening offers the opportunity to identify couples at risk of having a child with a genetic disease
- Screening for heterozygotes biochemical abnormalities
  - Specific molecular diagnostic tests (ASO array tests) or mutation analysis
  - Biochemical testing on tissue detects 50%, no need to know specific mutation

Challenges for a carrier screening program

- **Allelic heterogeneity**
- For cystic fibrosis, the DNA mutation panel can identify 25 of the most common mutations in the US population
- A person who has been identified as a "non-carrier" – has the **residual risk** of being a carrier of an infrequent mutant allele, that is not included in the panel

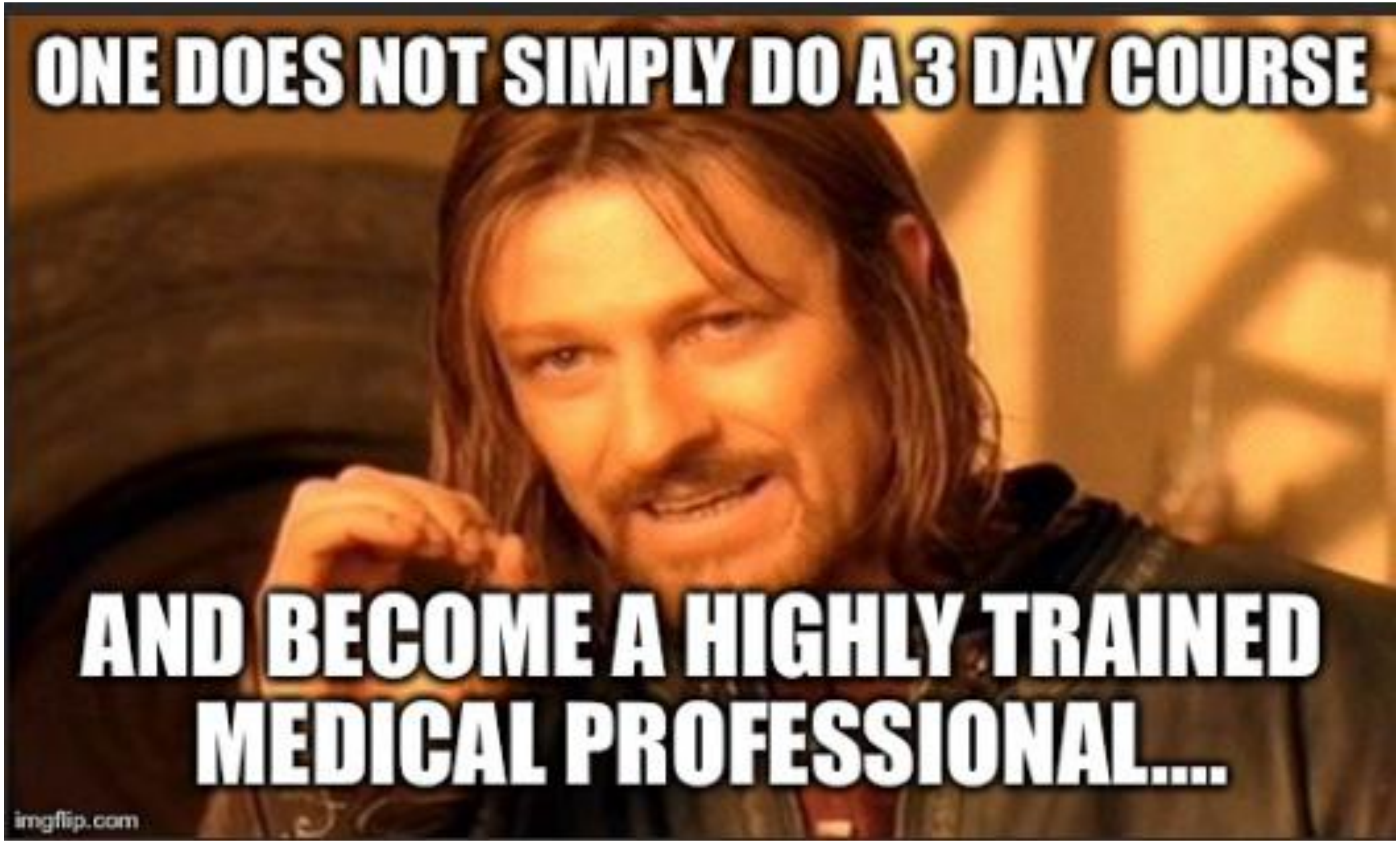
Carrier screening

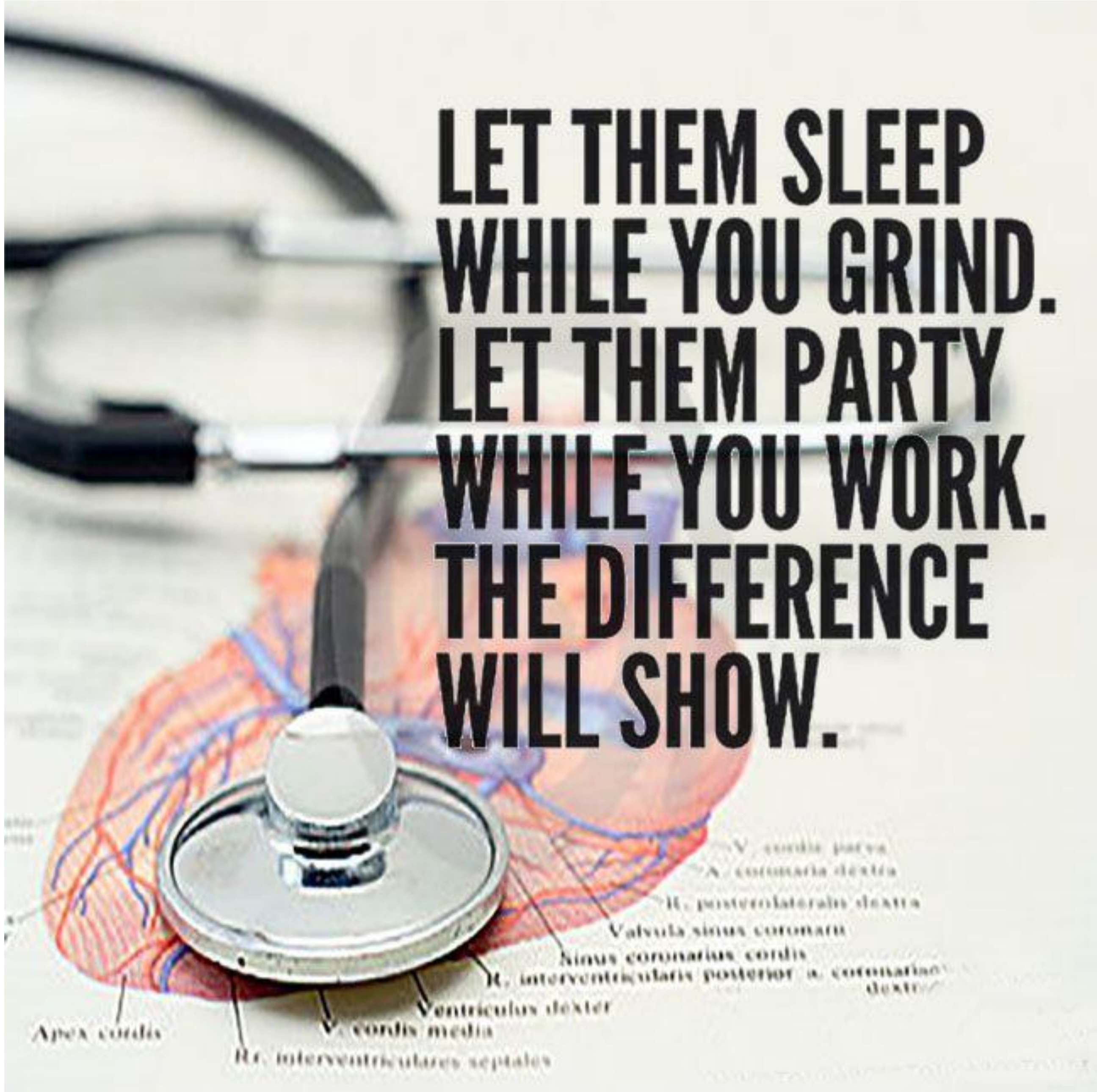
- Options for a couple in which both partners are carriers
  - Choosing not to have biological children
  - Artificial insemination or egg donation
  - Prenatal diagnosis and termination of an affected pregnancy
  - Prenatal diagnosis and planning for the care of an affected child
  - No prenatal testing

Ethical and legal issues in Medical Genetics "GENETHICS"

- The most common issues involve:
  - **Autonomy:** client has the right to make own decisions
    - Complications include patient age or economic status
    - What to do when asked "what would you do?" Avoid answering!
  - **Informed consent:** knowledge of all possible consequences
  - **The right to know or not know:** Late onset diseases?
  - **Coercion:** is the client being persuaded by family or friends?
  - **Confidentiality:** no release of data to third parties including relatives
    - Insurance?
  - **Reproductive decision making:** DMD is tested but not Wilson's??
    - Sex selection???
  - **Testing of minors or mentally challenged persons:** controversial,
    - Will the patient benefit from the test?
    - Is there a treatment or intervention?
    - Early onset vs. late onset diseases?

Clicker Questions Next!




LET THEM SLEEP
WHILE YOU GRIND.
LET THEM PARTY
WHILE YOU WORK.
THE DIFFERENCE
WILL SHOW.

The End